



TAMPINES JUNIOR COLLEGE

Preliminary Examinations 2008

CHEMISTRY (Higher 1)

8872/1

PAPER 1 Multiple Choice

Wednesday

27 August 2008

50 minutes

Additional materials: Multiple Choice Answer Sheet
Data Booklet

INSTRUCTIONS TO CANDIDATES

Write in soft pencil.

Write your name and Civics Group on the Answer Sheet in the spaces provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the one you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **8** printed pages.

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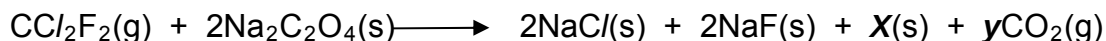
Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 How many molecules are there in 600 cm^3 of chlorine gas under room conditions?

A 1.50×10^{22} **B** 1.61×10^{22} **C** 3.00×10^{22} **D** 3.21×10^{22}

- 2 A process was devised to pass CCl_2F_2 through a packed bed of sodium ethanedioate to convert the CCl_2F_2 into solid and gaseous products.



What is the identity of the product **X** and the number **y**?

	X	y
A	C	2
B	C	4
C	2C	2
D	2C	4

- 3 Compound **Z** is a very unstable oxidising agent and will decompose violently at room temperature. Given that **Z** contains chlorine in an oxidation state of +5, which of the following could be **Z**?

A NCl_3 **B** NH_4Cl **C** NH_4ClO_3 **D** NH_4ClO_4

- 4 Which one of the following Period III elements has the same number of neutrons as $^{32}_{16}\text{S}$?

A $^{23}_{11}\text{Na}$ **B** $^{24}_{12}\text{Mg}$ **C** $^{28}_{14}\text{Si}$ **D** $^{31}_{15}\text{P}$

- 5 Which of the following particles has the same electronic configuration as the chloride ion, Cl^- ?

A F^- **B** Ne **C** Na^+ **D** Ca^{2+}

- 6 Silicon dioxide, SiO_2 , is an inert material with a high melting point. It is also extensively used in the manufacture of glass and bricks. Which type of structure explains these properties?

A A simple covalent structure with covalent bonds between silicon and oxygen atoms.
B A layered structure with covalent bonds between silicon and oxygen atoms and weak van der Waals' forces between the layers.
C A giant covalent structure with strong covalent bonds between silicon and oxygen atoms forming a 3-dimensional network.
D A giant ionic lattice with strong ionic bonds holding the oppositely charged ions together.

- 7 Which of the following molecules would you expect **NOT** to have a bond angle of 120° ?
A BCl_3 **B** SO_3 **C** $\text{H}_2\text{C}=\text{O}$ **D** PCl_3
- 8 The boiling point of iodine monochloride, ICl , is nearly 40°C higher than that of bromine, although the two substances have almost the same relative molecular mass.
 Which one of the following is the best explanation of this observation?
A ICl is a polar molecule.
B ICl has an ionic structure.
C The $\text{I} - \text{Cl}$ bond is stronger than the $\text{Br} - \text{Br}$ bond.
D The ionisation energy of iodine is lower than that of bromine.
- 9 Which molecule will **not** form hydrogen bond with another molecule of its own?
A CH_3CHO **C** NH_3
B CH_3NH_2 **D** $\text{CH}_3\text{CH}_2\text{OH}$
- 10 Which of the equations correctly defines the enthalpy change of formation of a compound?
A $\text{Na(s)} + \text{Cl(g)} \rightarrow \text{NaCl(s)}$
B $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O(l)}$
C $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$
D $\text{K(s)} + \text{Mn(s)} + 2\text{O}_2\text{(g)} \rightarrow \text{KMnO}_4\text{(l)}$
- 11 The radius and charge of each of six ions are shown in the table.

ion	A^+	B^+	C^{2+}	X^-	Y^-	Z^{2-}
radius / nm	0.15	0.19	0.16	0.15	0.19	0.16

The ionic solids AX , BY and CZ are of the same lattice type. What is the correct order of their lattice energies placing the one with the highest numerical value first?

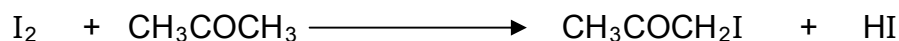
- A** $\text{AX} > \text{CZ} > \text{BY}$
B $\text{AX} > \text{BY} > \text{CZ}$
C $\text{CZ} > \text{BY} > \text{AX}$
D $\text{CZ} > \text{AX} > \text{BY}$
- 12 The experimental results obtained for the reaction between X and Y (at constant temperature) are given in the table below.

Expt	$[\text{X}] / \text{mol dm}^{-3}$	$[\text{Y}] / \text{mol dm}^{-3}$	Rate / $\text{mol dm}^{-3} \text{s}^{-1}$
1	0.1	0.1	0.004
2	0.1	0.2	0.016
3	0.2	0.1	0.008
4	0.2	0.2	a

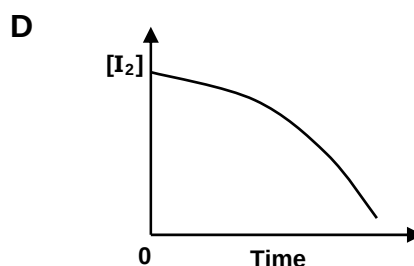
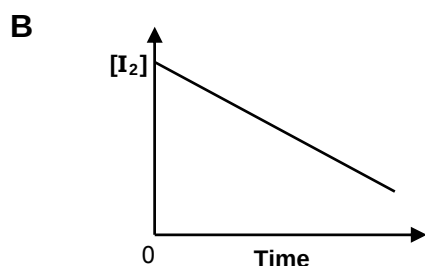
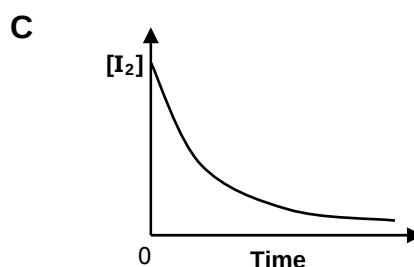
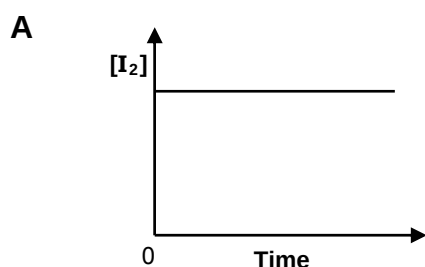
What is the value of **a**?

- A** 0.008 **B** 0.016 **C** 0.032 **D** 0.064

13 Iodine reacts with propanone according to the equation



The reaction of iodine with propanone is found to be zero order with respect to iodine. Which graph correctly show how the $[\text{I}_2]$ changes with time?

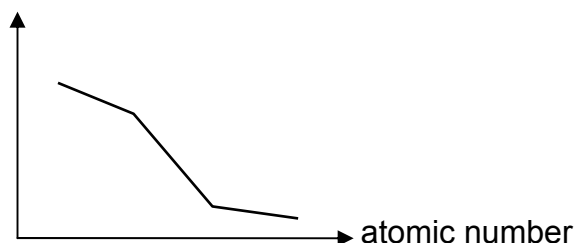


14 Given the following equilibrium: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ $\Delta H < 0$

Which of the following changes would affect both the proportion of ammonia present at equilibrium and the value of equilibrium constant, K_c ?

- A** adding a catalyst
- B** adding more nitrogen
- C** increasing the temperature
- D** increasing the total pressure

15 Which property when plotted against increasing atomic number gives the shape of the following graph?



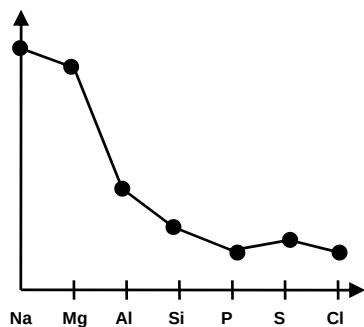
- A** ionic radius of Mg, Al, Si and P
- B** first ionisation energy of Mg, Ca, Sr and Ba
- C** boiling point of chlorides of Na, Mg, Al and Si
- D** electronegativity of Si, P, S and Cl.

16 An element from the third period is heated in chlorine. When the purified product is added to water, the resulting solution is found to be neutral. Which of the following could be the element?

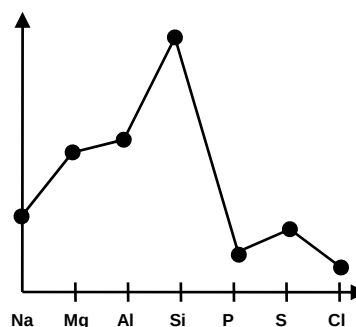
- A** sodium
- B** aluminium
- C** silicon
- D** phosphorus

17 Which of the following graph best represents the variation in melting points of the third period elements (Na to Cl)??

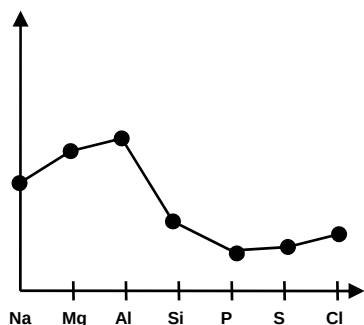
A



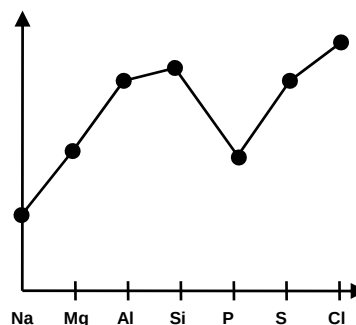
C



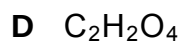
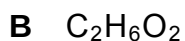
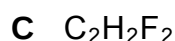
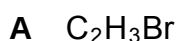
B



D



18 Which of the following formulae could represent a compound which can exhibit cis-trans isomerism?



19 The C_2H_2 molecule is linear.

What can be deduced from this about the numbers of σ and π bonds present in the molecule?

	σ	π
A	2	2
B	2	3
C	3	2
D	4	1

20 Which of the following statements best describes the reaction between ethene and hydrogen bromide?

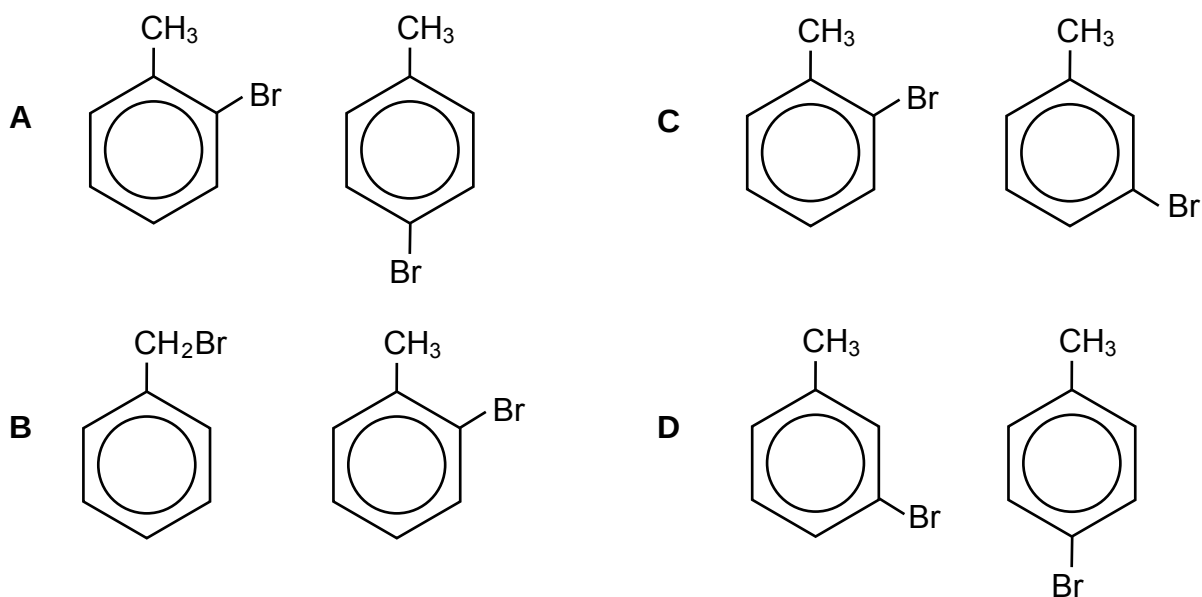
A Hydrogen bromide is decolourised.

B Bromoethane is the only product formed.

C 1,2-dibromoethane is the only product formed.

D A mixture of *cis*- and *trans*-1,2-dibromoethane is formed.

- 21 When methylbenzene is treated with bromine in the presence of a catalyst, a mixture of two monobromo isomers is formed.
What are the structure of these two isomers?



- 22 Which of the following statements best explain why fluoroalkanes are the least reactive halogenoalkanes?

- A The F^- ion is the most stable halide ion.
B Fluorine is much more electronegative than carbon.
C The C–F bond is the strongest carbon – halogen bond.
D The C–F bond is the most polar carbon – halogen bond.

- 23 Which of the following reagents can be used to distinguish between propan–1–ol and propan–2–ol?

- A alkaline aqueous iodine
B concentrated H_2SO_4
C lithium aluminium hydride
D sodium metal

- 24 Which statement about ethanal and propanone is **incorrect**?

- A Both may be prepared by oxidation of an alcohol.
B Both change the colour of warm acidified potassium dichromate (VI) from orange to green.
C Both react with 2,4-dinitrophenylhydrazine.
D Both give a positive tri-iodomethane (iodoform) test.

- 25 The carboxylic acid is obtained either by the hydrolysis of a nitrile **X** or by oxidation of an alcohol **Y**. Which of the following pairs could be **X** and **Y**?

- | | X | Y |
|---|--------------------|------------------------|
| A | CH_3CH_2CN | CH_3CH_2OH |
| B | $(CH_3)_2CHCN$ | $(CH_3)_2CHOH$ |
| C | $C_6H_5CH(CH_3)CN$ | $C_6H_5CH_2CH(OH)CH_3$ |
| D | $C_6H_5CH_2CN$ | $C_6H_5CH_2CH_2OH$ |

Section B

For each of the questions in this section one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26 In an experiment, the complete combustion of a hydrocarbon yields 3.52 g of carbon dioxide and 1.44 g of water. Assume that all measurements were obtained at same temperature and pressure.

Which of the following molecular formulae would fit these data?

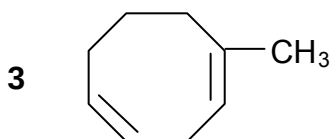
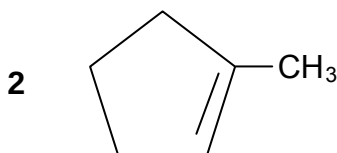
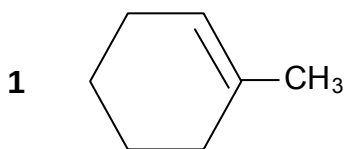


- 27 Which of the following pairs shows that the first compound has a higher melting point than the second?

- 1 potassium fluoride and potassium bromide
- 2 ammonia (NH_3) and phosphine (PH_3)
- 3 *trans*-but-2-ene and *cis*-but-2-ene

- 28 A hydrocarbon, X, on heating with an excess of hot concentrated acidified potassium manganate(VII), produces $\text{HO}_2\text{CCH}_2\text{CH}_2\text{CH}_2\text{COCH}_3$.

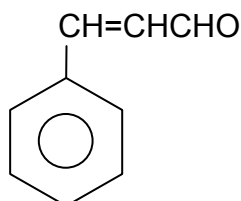
What could X be?



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A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

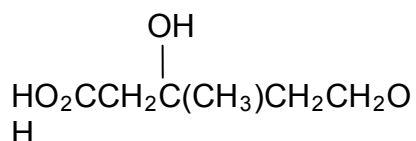
- 29** Cinnamaldehyde is an essential oil with the distinctive odour of cinnamon. It is used in some perfumes of natural, sweet, or fruity scents.



Which of the following statements about cinnamaldehyde is correct?

- 1 It exhibits cis-trans isomerism.
- 2 It decolourises aqueous bromine.
- 3 It reacts with Fehling's reagent.

- 30** Mevalonic acid is an intermediate in the biosynthesis of cholesterol. The acid has a number of functional groups related to this activity.



Which of the following properties does mevalonic acid have?

- 1 It can be esterified both by ethanoic acid and by ethanol.
- 2 It produces a yellow precipitate with warm alkaline I_2 (aq).
- 3 It contains both primary and secondary alcohol functional groups.

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